

Executive Summary

Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

Overview

This project presents a comprehensive product analytics and revenue optimization study for Afficionado Coffee Roasters using transaction-level sales data. The analysis identified high-performing products, evaluated category-level revenue dependence, and uncovered operational opportunities for menu optimization and profitability enhancement.

Business Problem

Afficionado Coffee Roasters faced challenges including high-volume products generating low revenue, premium products with low sales frequency, overcrowded menus, and limited visibility into product-level profitability and category dependency.

Project Objectives

The project aimed to identify top-selling and least-selling products, measure revenue contribution by product and category, analyze revenue concentration across the menu, and support menu simplification and operational efficiency.

Analytical Approach

The workflow included data cleaning, revenue computation, exploratory data analysis, product popularity analysis, revenue contribution analysis, Pareto analysis, KPI construction, and Streamlit dashboard development.

Key Findings

The analysis revealed strong Pareto behavior where a small percentage of products generated the majority of revenue. Long-tail products contributed minimally while increasing operational complexity. Category-level revenue dependency risks were also identified.

Strategic Recommendations

Recommendations include menu optimization, removal or redesign of low-performing products, promotion of high-margin products, inventory optimization, and operational simplification.

Business Impact

The project supports improved revenue optimization, enhanced operational efficiency, better menu engineering decisions, reduced inventory waste, and stronger long-term profitability.

Conclusion

This project demonstrates how transaction-level analytics can transform product strategy and revenue optimization within specialty coffee retail operations. The findings provide a foundation for scalable, data-driven business decision-making.

Key Deliverables

Deliverable	Status
EDA Notebooks	Completed
Revenue Contribution Analysis	Completed
Pareto Analysis	Completed
KPI Dashboard	Completed
Streamlit Dashboard	Completed
Research Paper Structure	Completed
Executive Summary	Completed